INTERNATIONAL WFI NUCLEAR PRODUCTS
Manufacturer of Piping and Pressure Vessel Components
4404 Haygood St - Houston, TX 77022
Phone: 713-695-3633 Fax: 713-695-3528
A Bonney Forge Company

Sold To: STUART C. IRBY COMPANY
1 CAPE MAY STREET
UNIT G
HARRISON NJ 07029-2402
United States

MTR #: 458,199
PO #: S014647269
Sales Order #: C002606743
Date: 07/21/2026

This product has not come in direct contact with mercury or any of its compounds, nor with any mercury-containing device employing a single boundary of containment. No welding performed.

We certify that the contents of this report are correct and accurate, and that all test results and operations performed by WFI or its subcontractors are in compliance with the material specification and requirements of the referenced code or standard, and that the material conforms to the dimensional requirements of the order. This document is in accordance with EN10204 3.1.

Certified Material Test Report

Heat Code: 8349AD4
Material: A694 16 F65

Item	Quantity	Description
1	1	16 (.500) X 2 (2.375ID) ST BORE SWP A694F65 BORED TO PASS 2 IN PIPE CE MAT# ITEM 23 PER ATTACHED DWG

Chemical Composition

	Al	B	C	Cr	Cu	Mn	Mo	N	Nb	Ni	P	S
Ladle	0.031	0.0002	0.150	0.12	0.23	1.200	0.040	0.014	0.0000	0.060	0.008	0.0040
	Si	Ti	V									
	0.170	0.000	0.08									
	Carbon Equivalency (C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15):				0.42							
Product	Al	B	C	Cr	Cu	Mn	Mo	N	Nb	Ni	P	S
	0.037	<0.0002	0.154	0.13	0.23	1.190	0.034	0.013	0.0026	0.068	0.005	0.0035
	Si	Ti	V									
	0.168	0.001	0.06									
	Carbon Equivalency (C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15):				0.42							

	Tensile PSI	Yield PSI	Elong (4D) %	RA %	Hardness
Product	85,200	66,800	31.60	73.30	162 BHN 162 BHN

Charpy V-Notch Properties

Specimen Size	Test Temperature	Ft Lbs	% Shear	MLE	Direction
10mm x 10mm	-50 F	107/101/93	80/75/70	82/76/62	Longitudinal

MILL/COUNTRY OF ORIGIN: STEEL DYNAMICS (USA) - HEAT #A181595
GRAIN SIZE IAW ASTM E112: #8.3

AUSTENITIZED BETWEEN 1675-1725°F FOR A MINIMUM OF 60 MINUTES PER INCH OF SECTION THICKNESS BUT NO LESS THAN 60 MINUTES AT TEMPERATURE (WATER QUENCHED); TEMPERED BETWEEN 1000-1030°F FOR 60 MINUTES OF SECTION THICKNESS MINIMUM (AIR COOLED).

Monica Rogers Quality Assurance Representative